JALU VEDA

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Profile

ML researcher and engineer (M.S. candidate, NTUST, expected 2027) building label-efficient and efficient-inference vision systems: self-supervised ViTs for medical imaging (published 2026), sign language recognition on event cameras and edge hardware, and AI surrogate modeling with TSMC. Deploys models with ONNX/TensorRT on Jetson-class devices. Background in embedded systems and industrial control.

Education

02/2025 – 02/2027
Taipei City, Taiwan

Master of Electronics and Computer Engineering,
National Taiwan University of Science and Technology
- GPA 4.00/4.30 - Recipient of NTUST Full Scholarship - Mobilizing Information Technology Laboratory Member
- Thesis: Whole-Pipeline Measurement and Compression of Skeleton-Based Continuous Sign Language Recognition on Edge Devices

08/2018 – 08/2022
Surabaya, Indonesia

Bachelor of Electrical Engineering with Electronics Specialization,
Institut Teknologi Sepuluh Nopember
- GPA 3.61/4.00, cum laude graduate - Practicum assistant for circuits and electronics (50+ students); Industrial Electronics Laboratory assistant
- Thesis: Control and Monitoring System for Plant NPK Fertilization Using the ESP32 Microcontoller

Experience

02/2025 – Present
Taipei City, Taiwan

National Taiwan University of Science and Technology (NTUST),
Graduate Student Researcher
- NTUST × TSMC collaboration: developed AI surrogates predicting a wafer's 100-point thickness profile from a chamber recipe, trained on 1,000 synthetic samples (Latin hypercube) from a rotation-aware RBF/CVD physics model
- Built an interactive 3D digital twin (Three.js + FastAPI) serving real-time profile predictions
- Co-authored a published self-supervised learning study for chest X-ray disease detection: designed the domain-adaptation strategy (freezing 8 of 12 ViT blocks) and ran the ablations; stroke model reached 89.60% ± 1.39 accuracy, beating RAD-DINO with ~4× fewer parameters

08/2023 – 04/2024
Lombok Barat, Indonesia

PLN Indonesia Power PLTU Jeranjang, *Electrical and Instrumentation Maintenance - PLN Indonesia Power, PLTU Jeranjang (3×25 MW coal-fired plant)*
- Performed predictive, preventive and corrective maintenance on 100+ pieces of electrical and instrumentation equipment (motors, transformers, generators, protections, DCS) across 3 plant units
- Led a two-person team to design and commission closed-loop PID furnace pressure control on a HollySys MACS DCS, replacing manual induced-draft fan operation and holding pressure within the −140 to −40 Pa safety window

Projects

2026 – Present

M.S. Thesis: Whole-Pipeline Measurement and Compression of Skeleton-Based Continuous Sign Language Recognition on Edge Devices,
PyTorch, ONNX, TensorRT, NVIDIA Jetson Orin Nano, MSKA, RTMPose/DWPose

- Deployed every neural stage as a TensorRT engine and profiled the full pipeline (decode → pose → CTC) on a Jetson Orin Nano with per-stage energy attribution; the recognition head and CTC are only 9.5–23.3% of system energy, though published efficiency figures cover the head alone
- Showed a released MSKA head is not extractor-agnostic: dev WER rises from 21.14 to 57.70 on a different pose estimator, caused by unnormalized keypoint confidence; fine-tuning recovers 21.78
- Cut system energy 2.41× for +1.06 WER (FP16 front end, pose on every 2nd frame, no retraining; batch regime). An RGB-aware teacher gains 0.6–0.8 WER, isolated by a retrained control with 95% paired bootstrap intervals

2025 – Present

SSL-Pretrained Event-Camera Continuous Sign Language Recognition,
PyTorch, ViT-S/14, Generative Event Pretraining (GEP), EvSign, EvRealHands

- Testing whether a self-supervised, event-pretrained ViT-S/14 encoder transfers to continuous sign language recognition on EvSign, a task bottlenecked by only 5,570 labeled sequences
- Built the pipeline on the published VAC/CorrNet skeleton (event-to-frame → ViT → TCN+BiLSTM → CTC) and designed attribution-controlled experiments (frozen vs. fine-tuned, domain-adapted vs. baseline) with multi-seed reporting

2025

Face Recognition Attendance System, *Group project · Raspberry Pi 4, OpenCV, AWS (Rekognition, Lambda, S3, DynamoDB, SNS), React, Flask · github.com/ontherays/smartFaceRecognistionAWS*

- Built an end-to-end system: edge face detection on a Raspberry Pi 4 with AWS Rekognition for identity verification, a serverless backend for attendance logging and notifications, and a React/Flask dashboard

Publications

09/2026

DINOv2-Pretrained Vision Transformers for Stroke and Heart Failure Detection from Chest X-rays with Limited Labeled Data,
Journal of Imaging Informatics in Medicine (Springer Nature)

- Co-developed a three-stage SSL framework (DINOv2 pretraining → CXR domain adaptation → linear probing); designed the domain-adaptation strategy, partially freezing the ViT backbone (bottom 8 of 12 blocks) to preserve general visual features while adapting to CXR morphology

09/2022

Sistem Kontrol dan Monitoring Pemupukan NPK Tanaman dengan Mikrokontroler ESP32, *Jurnal Teknik ITS*

- ENG: Control and Monitoring System for Plant NPK Fertilization Using the ESP32 Microcontoller
- Published journal article derived from undergraduate thesis; designed a closed-loop PID fertilization system using ESP32, NPK soil sensors, and peristaltic pumps, with remote monitoring via Blynk

Skills

Machine Learning

- Self-supervised learning (DINOv2)
- Vision Transformers
- CTC sequence models
- knowledge distillation
- FP16/INT8 quantization
- XGBoost/LightGBM/CatBoost
- PCA/autoencoders
- SHAP

Computer Vision

Pose estimation (RTMPose/DWPose), event cameras, object detection, OpenCV

Tools and Deployment

Python, C/C++, PyTorch, scikit-learn, ONNX, TensorRT, Docker, Git, Linux, FastAPI, Jetson Orin Nano, Raspberry Pi, latency/energy profiling, AWS

Languages

English (IELTS 7.5), Mandarin (basic), Indonesian (native)